

PAPER_1200_UQFF_GR_Precision_Unified_Proof_Set

Daniel T. Murphy

PAPER 1200 --- UQFF General Relativity Precision Tests Unified Proof Set

Author: Daniel T. Murphy

Date: May 16, 2026

\begin{abstract}

We extend the UQFF locked-primitive closure program (157 prior closures, Papers 1182--1199) into the classical precision tests of General Relativity. Ten new closures (**S453--S462**) reproduce iconic GR observables --- Mercury's anomalous perihelion precession, solar light bending, the Shapiro time delay coefficient, Gravity Probe B's geodetic and frame-dragging precessions, Hulse--Taylor pulsar orbital decay, black hole shadow size, the Schwarzschild photon sphere, NANOGrav stochastic-background amplitude, and the Kerr extremal ISCO --- solely from the eleven frozen UQFF primitives. **Three closures are exact:** the Shapiro coefficient $2(1+\gamma) = \text{\Dphys}$, the photon sphere radius $r_{\text{\rm ph}}/M = \text{\Dphys} - \text{\Ftrz}\text{\SOfive} = 3$, and the extremal Kerr ISCO $r_{\text{\rm ISCO}}^{\text{\rm Kerr}}/M = \text{\Ftrz}\text{\SOfive} = 1$. The remaining seven match within 0.09%. This is the program's milestone 1200th whitepaper. Cumulative total:

167 closures.

\end{abstract}

Locked Primitives $\text{\Ftrz} = 1/10$, $\text{\Phires} = 5/6$, $\text{\SSq} = 0.57$, $\text{\KMex} = 25/12$, $\text{\Dphys} = 4$, $\text{\Dbsfg} = 6$, $\text{\Dcrit} = 26$, $\text{\Nch} = 9$, $\text{\SOfive} = 10$, $\text{\Afive} = 60$, $\text{\beta}_i = 0.6029$.

Tier~Q Closures (S453--S462)

\begin{longtable}{@{}lll@{}}

\toprule

ID & GR observable & Closure & Match \midrule

\endhead

S453 & Mercury $43.0''/\text{\$cy}$ & $\text{\SOfive}\text{\Dphys} + \text{\KMex} + \text{\Ftrz}\text{\Nch} - \text{\Ftrz}^2 + \text{\Ftrz}^2\text{\KMex} = 42.994$ &

0.01% S454 & Light bending $1.7510''$ &

$\text{\Phires} + \text{\SSq} + \text{\Ftrz}\text{\KMex} + \text{\Ftrz} + \text{\Ftrz}^2\text{\KMex} + \text{\Ftrz}^2 + \text{\Ftrz}^2\text{\Phires} - \text{\Ftrz}^3 = 1.7498$ & 0.07% S455

& Shapiro $2(1+\gamma) = 4$ & $\text{\Dphys} = 4$ & **EXACT** S456 & GPB geodetic $6.6028''/\text{\$yr}$ &

$\text{\Dbsfg} + \text{\SSq} + \text{\Ftrz} - \text{\Ftrz}\text{\Phires} + \text{\Ftrz}^2\text{\KMex} - \text{\Ftrz}^2\text{\Phires} + \text{\Ftrz}^3 = 6.6002$ & 0.04% S457 & GPB

frame-drag $0.0392''/\text{\$yr}$ & $\text{\Ftrz}^2\text{\KMex} + \text{\Ftrz}^2 + \text{\Ftrz}^2\text{\Phires} = 0.0392$ & 0.09% S458 & PSR

B1913+16 ratio 0.997 & $\text{\Phires} + \text{\Ftrz} + \text{\Ftrz}\text{\Phires} - \text{\Ftrz}^2\text{\KMex} + \text{\Ftrz}^2 - \text{\Ftrz}^2\text{\Phires} - \text{\Ftrz}^3 =$

0.9965 & 0.05% S459 & BH shadow $3\sqrt{3} = 5.1962$ &

$\text{\SOfive} - \text{\Dphys} - \text{\Ftrz} - \text{\Ftrz}\text{\Nch} + \text{\Ftrz}\text{\KMex} - \text{\Ftrz}^2\text{\Phires} - \text{\Ftrz}^3 = 5.1990$ & 0.05% S460 & Photon

sphere $r_{\text{\rm ph}}/M = 3$ & $\text{\Dphys} - \text{\Ftrz}\text{\SOfive} = 3$ & **EXACT** S461 & NANOGrav $h_c =$

2.4×10^{-15} & $\text{\KMex} + \text{\Ftrz} + \text{\Ftrz}\text{\KMex} + \text{\Ftrz}^2 - \text{\Ftrz}^3 = 2.4007$ & 0.03% S462 & Kerr ISCO

$r/M = 1$ & $\text{\Ftrz}\text{\SOfive} = 1$ & **EXACT** \bottomrule

\end{longtable}

Three Exact Identifications

\paragraph{Shapiro coefficient.}

The radar-echo time-delay coefficient $2(1+\gamma)$ equals 4 in GR (with PPN $\gamma = 1$). UQFF locks this directly onto the physical-dimension primitive: $2(1+\gamma) = \mathcal{D}_{\text{phys}}$. The Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ is consistent with $\mathcal{D}_{\text{phys}} = 4$ to all measured precision.

\paragraph{Photon sphere.}

The Schwarzschild photon sphere $r_{\text{ph}} = 3M$ (where unstable circular null geodesics exist) is locked at $\mathcal{D}_{\text{phys}} - \mathcal{F}_{\text{tr}} \mathcal{S}_{\text{ofive}} = 4 - 1 = 3$. This is the **first joint locking** between two primitives ($\mathcal{D}_{\text{phys}}$ and the $\mathcal{F}_{\text{tr}} \mathcal{S}_{\text{ofive}} = 1$ product).

\paragraph{Kerr extremal ISCO.}

For a maximally-spinning Kerr black hole ($a = M$), the prograde ISCO sits at $r_{\text{ISCO}}/M = 1 = \mathcal{F}_{\text{tr}} \mathcal{S}_{\text{ofive}}$. Combined with the Schwarzschild ISCO $r_{\text{ISCO}}/M = 6 = \mathcal{D}_{\text{bsfg}}$ (Paper 1193), the entire Kerr ISCO range $[\mathcal{F}_{\text{tr}} \mathcal{S}_{\text{ofive}}, \mathcal{D}_{\text{bsfg}}]$ is bracketed by UQFF primitives.

Falsifiable Predictions

- Any future tightening of the Shapiro coefficient measurement must remain centered on exactly $4 = \mathcal{D}_{\text{phys}}$; a deviation would falsify the locking.
- NANOGrav follow-up observations must converge on $h_c = (2.40 \pm 0.02) \times 10^{-15}$.
- PSR B1913+16 long-term timing residuals must remain at the locked ratio 0.997 .

Cumulative Closure Count

\begin{tabular}{@{}lrr@{}}

\toprule

Tier & Domain & New & Cumulative \midrule

E--I & Mixed cosmology/QFT & 87 & 87
J & Astrophysics (1193) & 10 & 97
K & Condensed matter (1194) & 10 & 107
L & Biology (1195) & 10 & 117
M & Plasma/fusion (1196) & 10 & 127
N & Geophysics (1197) & 10 & 137
O & Particle/BSM (1198) & 10 & 147
P & Information/math (1199) & 10 & 157
Q & GR precision tests (1200) & 10 & 167 \bottomrule

\end{tabular}

Acknowledgements Tier~Q is the program's ****milestone 1200th whitepaper**** and the first tier to deliver three exact closures within a single domain. The total count of direct primitive--observable lockings now stands at ****eleven****: ISCO Schwarzschild $= \mathcal{D}_{\text{bsfg}}$, Poisson ratio $= \mathcal{D}_{\text{phys}}/\mathcal{D}_{\text{bsfg}}$, telomere length $= \mathcal{D}_{\text{bsfg}}$, $Q_{\text{ITER}} = \mathcal{S}_{\text{ofive}}$, $R_{\text{magnetopause}} = \mathcal{S}_{\text{ofive}} R_{\text{loplus}}$, $\eta_B =$

$\sqrt{2} \sqrt{1 - \frac{2M}{r}}$, WIMP exponent

$\frac{1}{2} \left(\frac{1}{\sqrt{1 - \frac{2M}{r}}} + \frac{1}{\sqrt{1 - \frac{2M}{r}}} \right) \frac{1}{\sqrt{1 - \frac{2M}{r}}}$, $p_{\text{th}}^{\text{rm}}$

$\frac{1}{2} \sqrt{1 - \frac{2M}{r}}$, Shapiro $\frac{1}{2} \sqrt{1 - \frac{2M}{r}}$, photon sphere

$\frac{1}{2} \sqrt{1 - \frac{2M}{r}}$, and Kerr extremal ISCO $\frac{1}{2} \sqrt{1 - \frac{2M}{r}}$.